

Smart Metro Booking System

Project Planning Phase

Date

Team ID

Project Name: Smart Metro Booking System

Maximum Marks: 5 Marks

1. Project Planning Phase

1. Purpose

This document lays out the project planning approach for digitizing metro ticket booking on ServiceNow. It turns the solution requirements and technology stack into a structured plan — covering scope, work breakdown, timeline, roles, and risk management — so the build proceeds in a controlled, milestone-driven way.

2. Project Scope & Deliverables

The planning phase covers full delivery of the ServiceNow-based metro ticketing solution, from catalog configuration through go-live. Key deliverables include:

- A configured Service Catalog item with Source, Destination, Passenger Type, and Number of Tickets variables.
- Automated fare calculation logic, including concessions for Student / Senior Citizen categories.
- QR code generation and validation integrated with the booking and payment flow.
- Station Details Mapping covering routes, distances, and fare slabs.
- Passenger-facing UI pages across Mobile App, Portal, and WhatsApp channels.
- A tested, production-ready solution following UAT sign-off.

3. Work Breakdown Structure

The project splits into four major work streams — Requirements & Design, Development, Integration, and Testing & Deployment — each further divided into concrete work packages aligned to the project milestones.

Figure 1 — Work Breakdown Structure for Metro Digital Ticketing

4. Project Timeline & Milestones

Delivery follows six sequential milestones. The timeline below shows planned durations in project weeks, with Fare Logic Script and Station Details Mapping running partly in parallel with Catalog Form Setup to compress the overall schedule.

Figure 2 — Indicative Project Timeline (in weeks)

4.1 Milestone Details

Milestone	Description	Owner
Catalog Form Setup	Configure Service Catalog item with booking variables and validations.	IT Admin / ServiceNow Dev
Fare Logic Script	Build fare calculation business rules and concession logic.	ServiceNow Developer
QR Code Integration	Integrate QR generation API into the post-payment workflow.	Integration Developer
Station Details Mapping	Set up station master data, routes, and fare slabs.	Metro Operations Team
UI Pages	Build passenger-facing booking and ticket display screens.	UI/UX Developer

Milestone	Description	Owner
UAT & Deployment	User acceptance testing and production rollout.	QA Team / Project Manager

5. Roles & Responsibilities (RACI)

The RACI matrix below spells out who is Responsible, Accountable, Consulted, and Informed for each major milestone.

Milestone	Passengers	Station Mgr	Ops Team	IT Admin
Catalog Form Setup	I	I	C	R/A
Fare Logic Script	I	I	C	R/A
QR Code Integration	I	I	I	R/A
Station Details Mapping	I	C	R/A	C
UI Pages	C	I	I	R/A
UAT & Deployment	C	C	A	R

R = Responsible, A = Accountable, C = Consulted, I = Informed.

6. Risk Management Plan

Risk	Impact	Mitigation
Delay in payment gateway API access/credentials	High	Initiate vendor onboarding in parallel with Catalog Form Setup.
Incomplete or inaccurate station master data	Medium	Validate station and fare data with Operations Team before Fare Logic Script begins.
QR code duplication or security gaps	High	Use signed/encrypted QR tokens and test tamper scenarios during UAT.
Low passenger adoption of digital channels	Medium	Run a pilot at select stations with in-person support before full rollout.
AFC gate hardware compatibility issues	Medium	Conduct early hardware integration testing during the QR Code Integration milestone.

7. Assumptions & Constraints

- ServiceNow instance and required licenses (Catalog, Flow Designer, Integration Hub) are provisioned before the planning phase starts.
- Payment gateway and QR generation vendors are finalized and API access is available.
- Station Master Data is owned and maintained by the Metro Operations Team.
- UAT will be conducted with representative passengers, Station Managers, and IT Admins.
- Go-live is contingent on successful UAT sign-off from all stakeholder groups.